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Assessment of public attitude and uptake experience of COVID-19 tests in the United Kingdom and China: a comparative infodemiology study

Zhiyuan Hou, Leesa Lin, Yi Song, Qian Wang, Jialu Pu, Fiona Sun, Yixuan Zhang, Xinyu Zhou, Heidi Larson

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Assessment of public attitude and uptake experience of COVID-19 tests in the United Kingdom and China: a comparative infodemiology study

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Abstract

Background: Massive community-wide testing has become a cornerstone of management strategies for the COVID-19 pandemic. This study aimed to assess public attitudes, risk perception, information needs, perceived access, and uptake experience about COVID-19 testing in the UK and China.

Objective: This study aimed to assess public attitudes, risk perception, information needs, perceived access, and uptake experience about COVID-19 testing in the UK and China.

Methods: We collected and manually coded 3856 UK tweets and 9299 Chinese Sina Weibo posts mentioning COVID-19 testing from June to July, 2020. We adopted a grounded theory framework to conduct content analysis. Descriptive analysis, time trends, geographical mapping, and chi-squared test were performed to assess the temporal, spatial, and socio-demographic characteristics that determine the difference of attitudes or uptake of COVID-19 tests.

Results: Social media users in the UK demonstrated a higher percentage of support towards COVID-19 testing than those in China; comparatively, there were much wider reports of public uptake of COVID-19 tests in mainland China than the UK, and uncomfortable experiences and logistical barriers to testing were more expressed in China. The top reasons for taking COVID-19 tests were personal health needs, community-wide testing, and mandatory testing policies for travel, with major differences in the ranking order between two countries. Rumours and information need about COVID-19 testing were also identified.

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The local epidemic situation influenced public attitudes towards individual COVID-19 testing and mass community-wide testing. Policies and information campaigns that emphasise the importance of timely testing and provide a supporting environment for accessing a test are key to normalise COVID-19 testing as a part of life during the pandemic and to increase uptake.

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Massive community-wide testing has become a cornerstone of management strategies for the COVID-19 pandemic. This study aimed to assess public attitudes, risk perception, information needs, perceived access, and uptake experience about COVID-19 testing in the UK and China.

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Results

Social media users in the UK demonstrated a higher percentage of support towards COVID-19 testing than those in China; comparatively, there were much wider reports of public uptake of COVID-19 tests in mainland China than the UK, and uncomfortable experiences and logistical barriers to testing were more expressed in China. The top reasons for taking COVID-19 tests were personal health needs, community-wide testing, and mandatory testing policies for travel, with major differences in the ranking order between two countries. Rumours and information need about COVID-19 testing were also identified.

Conclusion

The local epidemic situation influenced public attitudes towards individual COVID-19 testing and mass community-wide testing. Policies and information campaigns that emphasise the importance of timely testing and provide a supporting environment for accessing a test are key to normalise COVID-19 testing as a part of life during the pandemic and to increase uptake.

Key words: COVID-19, test, public response, sentiment, social media, UK, China

Introduction

On March 11th, 2020, the World Health Organization (WHO) declared COVID-19 as a pandemic(1), and many countries initiated lockdowns as an acute response to the pandemic. With an understanding of how the virus progresses over time, testing kits for COVID-19 were developed and test-trace-isolate for those at risk has become a regular mitigation measure. Massive community-wide testing helped public health investigators understand the prevalence, contagiousness, and mortality of the disease(2), and has made it possible for communities to exit lockdowns and rapidly control potential resurgences while still awaiting a safe vaccine. As of December 15th, over 1 billion COVID-19 tests have been processed globally, and accordingly 71,583,891 cases have been confirmed(1, 3).

As of December 16th, the UK had performed the most COVID-19 tests per capita and China had performed the second most cumulative tests, roughly 160 million(3) – the governments swiftly secured test kits, reviewed regulations, and mobilised manpower for the introduction of COVID-19 tests, which required tremendous resources and coordination with multiple entities. Due to a lack of preparation and early strategies, such as a pause on community testing in March, the UK suffered a testing insufficiency at the beginning of the epidemic(4). Following the establishment of National

Laboratory Network and later the launch of the National Health System (NHS) Test and Trace service on May 28, the situation has eased. People with COVID-19 symptoms could receive a test from the NHS for free, and since July, those engaged in high-risk jobs were also promised regularly testing by the UK government(5). Entering November, areas with high level alerts for COVID-19 in the UK began engaging in mass testing for people with and without symptoms(6). An emphasis on testing asymptomatic individuals, which was discouraged in the early stage to save resources, is increasingly encouraged by the government as testing capacity increases(7). In China, COVID-19 testing has been made available to the public, including asymptomatic individuals, and since May 2020, mass testing has been conducted as a standard procedure in places where new outbreaks surged(8). During the resurgence in Beijing in June, 3.56 million individuals at risk were quickly tracked and tested(9), and the resurgence in Beijing was brought under control within one month.

The powerful capacity of social media to offer real-time big data on public sentiment and opinions has made social listening an important tool to develop and assess risk communication strategies during crises. In recent years, social listening has become an increasingly popular approach for public sentiment and opinions analysis, especially during epidemics, such as 2009/2010 H1N1(10), 2013/2014 MERS(11) and 2014 Ebola(10). Existing studies have mainly focused on public attitudes and responses to non-pharmaceutical interventions such as lockdowns, social distancing, and mask wearing(12-15), and their findings helped to adapt control strategies in the early stage of COVID-19 epidemic. As COVID-19 testing is currently considered a critical response strategy, the public responses to testing strategies must be assessed. As such, this infodemiology study aimed to assess public attitudes, risk perception, information needs, perceived access, and uptake experience about COVID-19 testing in the UK and China. Findings from this study may help governments adapt their COVID-19 testing strategies.

Method

Data collection

We collected social media posts from the most popular social media platforms in the UK and China: tweets (the UK) and Sina Weibo posts (mainland China) mentioning COVID-19 testing from June 1 to July 15, after the launch of the NHS Test and Trace system in the UK and mass testing for COVID-19's resurgence in Beijing, China. The keywords used for collecting tweets or Weibo posts were "covid test", "covid19 test", "covid-19 test", "coronavirus test", "test for covid", and "test for coronavirus" (□□□□ in Chinese). Posts from news and organizational accounts, duplicate posts, and quotes without expressing users' comments were removed. After removing posts that did not meet inclusion criteria, a 10% random sample of all eligible posts was performed. In total, 3856 tweets from the UK, including 794 tweets from London, as well as 9299 Weibo posts from mainland China, including 3155 posts from Beijing, were included for formal analysis and coding.

Data analysis

We identified and classified posts which expressed personal opinion or discussions on COVID-19 testing. The content analysis of tweets and Weibo posts adopted a grounded theory framework by which themes emerged from the data(16). Two researchers firstly classified a subsample of 700 posts independently, and the final codebook was determined through discussion. This coding framework covered public attitudes, risk perception, information needs, perceived

access, and uptake experience about COVID-19 testing. Each Tweet or Weibo post was independently coded by two researchers; disagreements were resolved by discussion with a third-coder. Cohenκ(17) was used to measure intercoder reliability, which wasκ=0.825 in the final revision.

A descriptive analysis was performed to show the percentage of topics for both Weibo and Twitter data. We plotted the time trends of acceptance rate towards general COVID-19 test by week. Geographical distributions of post numbers and percentage of oppositional attitude among the UK and mainland China were plotted by regions or provinces. χ^2 test was used to determine the difference of attitudes or behaviours of COVID-19 by gender, age, and education.

Results

Daily new COVID-19 case numbers in the UK and China

As shown in Figure 1, from May 28 to July 15, daily new COVID-19 confirmed cases demonstrated a decreasing trend in the UK, and stabilized around 50 in London. With the decrease of new cases, the COVID-19 Alert Level in the UK was switched to Level 3 on June 19, representing a COVID-19 epidemic became in general circulation. On June 29, Leicester became the first city in the UK to undergo a local lockdown after a resurgence of cases. During the same time frame, daily new cases in China fluctuated under 60, and new cases in Beijing accounted for the majority of these cases. There were no more local confirmed cases in Beijing after July 6.

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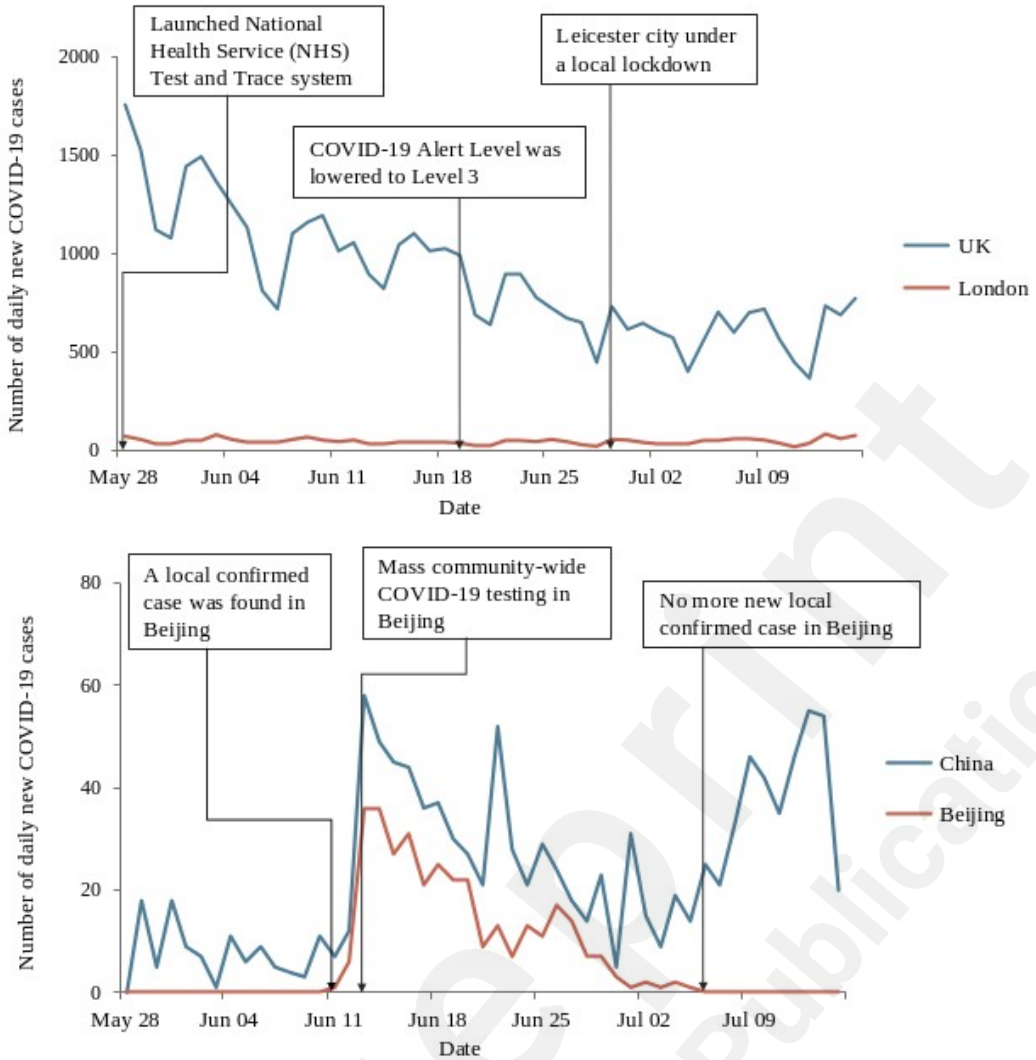

Figure 1.

Numbers of daily new COVID-19 cases in the United Kingdom (excluding London), London, mainland China (excluding Beijing), and Beijing from May 28 to July 15, 2020

Attitudes: Acceptance of COVID-19 test

In the UK, 64.6% (2390/3700) of tweets across the country and 69.2% (520/751) from London supported individual COVID-19 tests in general, while 4.7% (174/3700) across the country and 8.1% (61/751) from London opposed it, respectively. In China, 29.8% (2649/8879) Weibo posts across the country and 28.3% (848/2991) posts from Beijing supported COVID-19 tests in general, whereas 8.7% (776/8879) posts across the country and 10.2% (304/2991) posts from Beijing opposed it, respectively. Regarding government-led mass community-wide COVID-19 testing, the UK (1.1%, 23/2075) and London (1.1%, 6/530) showed less opposition than experienced across mainland China (4.8%, 76/1594) and Beijing (8.8%, 58/661). 2487 tweets from the UK with an oppositional attitude expressed discontent with governmental COVID-19 testing policies, including taxing on tests, voting against routine testing for front line workers, publishing wrong or untimely data, and other complaints. In China, Weibo posts with oppositional attitudes mainly regarded the trouble associated with having to obtain test results before travelling, visiting doctors, and so on. As for government-led mass community-wide COVID-19 testing, 31.0% (304/982) of Weibo posts from mainland China opposed the policies and/or questioned their effectiveness.

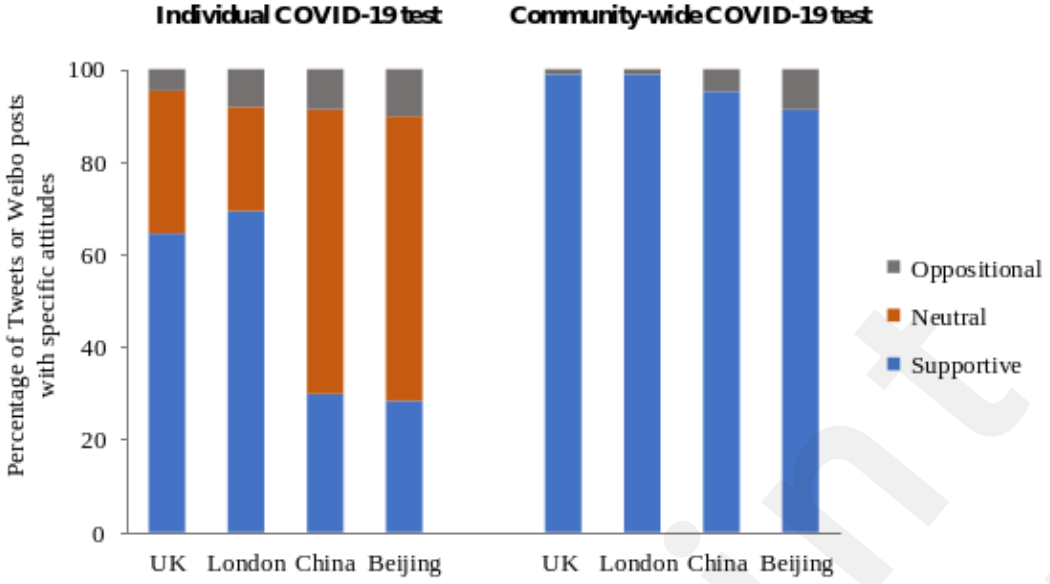

Figure 2. Percentage of tweets or Weibo posts with attitudes towards individual COVID-19 test and community-wide test from June 1 to July 15, 2020

Time trends analysis and Geographical mapping

Time trends analysis showed in the UK, posts with supportive attitudes towards individual COVID-19 tests first increased from 54% to 86%, then dropped to 39% after the enactment of The Health Protection (Coronavirus) Regulations in July (18), while tweets with oppositional attitudes increased from 2% to 16%. In China, Weibo posts showing supportive attitudes reduced from 54% to 7% during this period, while those with neutral attitudes increased from 34% to 86% and posts with oppositional attitudes stabilized around 10%. In Figure 4, London and Beijing had the largest number of posts on COVID-19 tests, and highest oppositional attitudes, compared with the rest of the countries.

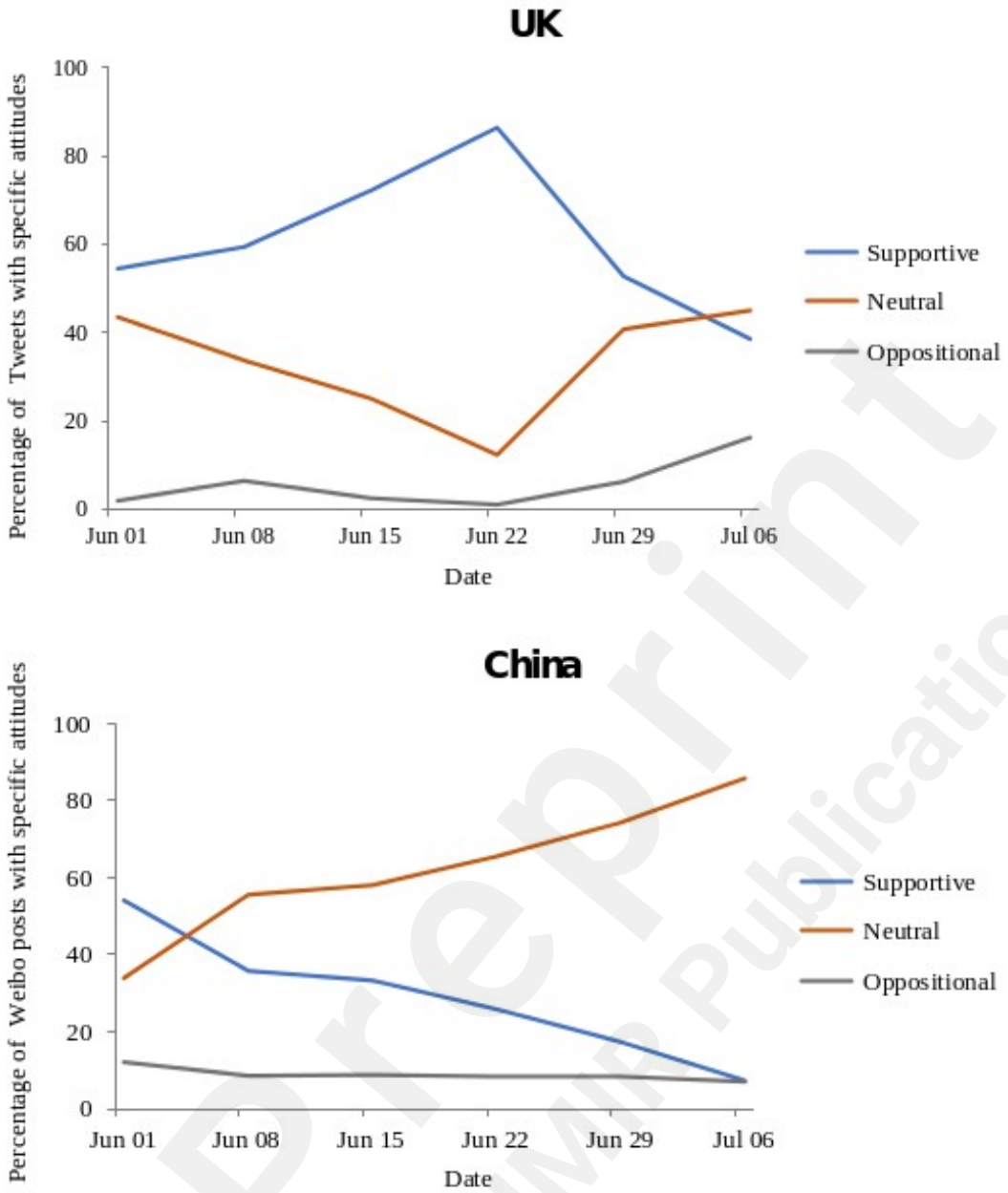

Figure 3. Percentage of tweets or Weibo posts with attitudes towards COVID-19 test by week from June 1 to July 15, 2020

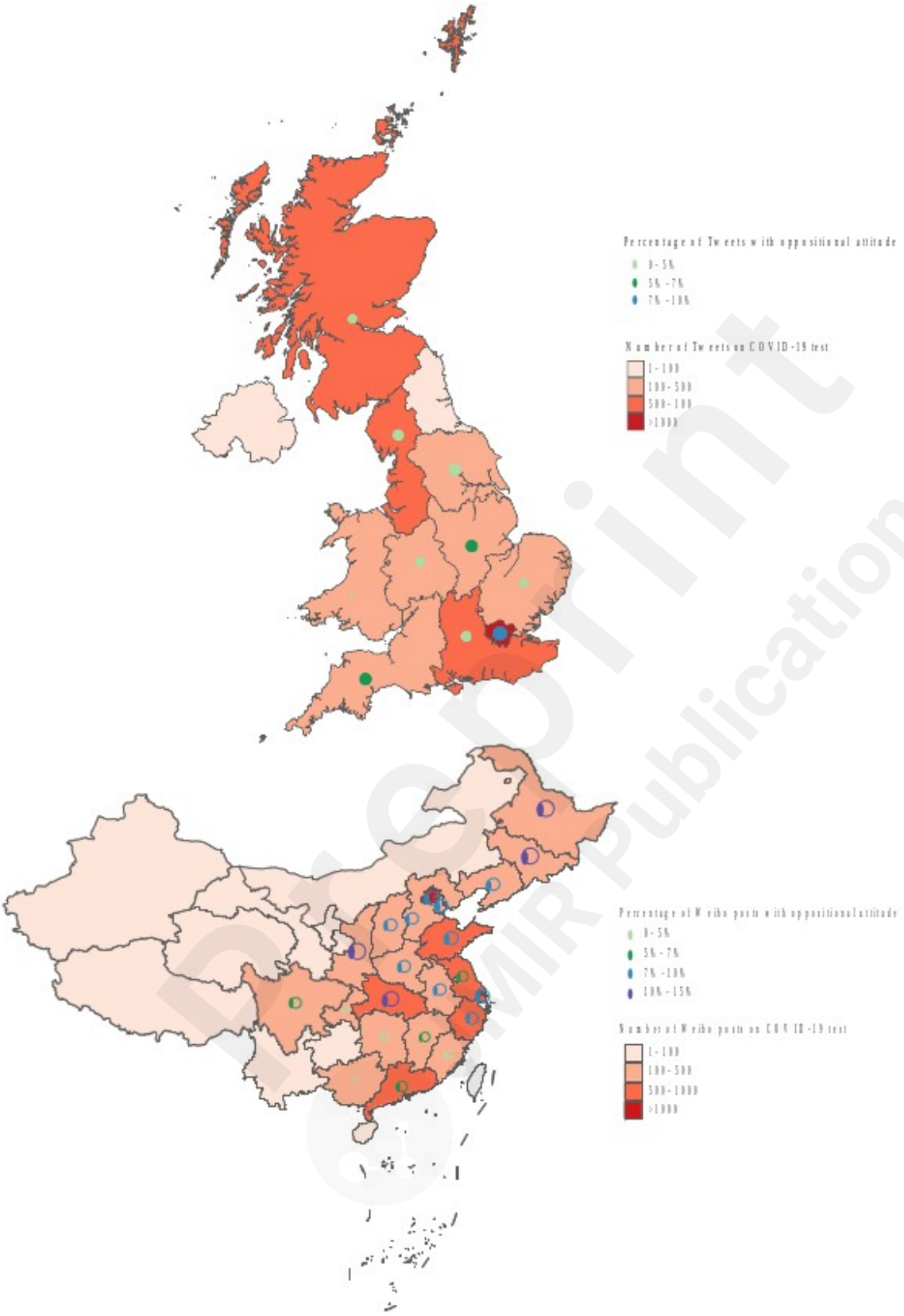

Figure 4. Percentage of tweets or Weibo posts with oppositional attitude towards COVID-19 test by geographical distribution in the United Kingdom and China from June 1 to July 15, 2020

Uptake of COVID-19 tests

Overall, 4.6% (178/3856) of tweets in the UK and 4.9% (39/794) in London reported intending to or having taken

COVID-19 tests. In the UK, rationales for taking tests included personal health needs due to possible exposure, symptoms or worry (43.0%, 37/86), mandatory testing policies for travel (34.9%, 30/86), and mass community-wide testing led by the government (22.1%, 19/86). Comparatively, there was a much wider report of public uptake of COVID-19 tests from China (35.7%, 3318/9299) and Beijing (46.3%, 1462/3155). A total of 1600 Weibo posts (17.2%, 1600/9299) from China reported the reasons for taking tests, including community-wide testing led by the government (49.0%, 784/1600), mandatory testing policies for travel (41.2%, 659/1600), and personal health needs (10.2%, 163/1600). There was much wider reporting of community-wide testing as the reason for taking tests from Beijing (65.9%, 543/824) than in China generally (49.0%, 784/1600).

Sentiment towards COVID-19 disease and tests

In both the UK and China, social media users expressed a high perceived risk of the COVID-19 pandemic (more than 95%). 4.3% (164/3856) of tweets across the UK and 6.2% (49/794) from London concerned the effectiveness of COVID-19 tests, respectively; of them, 20.7% (34/164) across the UK and 32.7% (16/49) from London expressed confidence in its effectiveness, and 79.3% (130/164) across the UK and 67.3% (33/49) from London doubted it, respectively. In China, 4.5% of posts (417/9299) across the country and 5.2% (164/3155) from Beijing concerned the tests' effectiveness respectively; of them, 65.9% (275/417) across China and 62.2% (102/164) from Beijing expressed confidence in its effectiveness, and 34.1% (142/417) across China and 37.8% (62/164) from Beijing doubted it, respectively.

Inquiries and rumours related to COVID-19 tests

Inquiries and rumours about COVID-19 tests could be found on both British and Chinese social media platforms (UK: 1.9%, 75/3856 and 1.4% 55/3856; China: 5.6%, 517/9299 and 2.1%, 192/9299). The most mentioned inquiries in the UK about COVID-19 test were "*how many people have received a test*", "*delay in sharing testing data with English councils*", and "*why they send our tests to the United States*", while in China, inquiries included "*whether tests are needed before travelling*" and "*how much it cost to take a test*". In the UK, rumours related to COVID-19 tests were about the motive and reliability of testing: "*the state earned money by COVID-19 test*" and "*COVID-19 test results were invalid*". Comparatively, in China, rumours included "*positive testing results for someone in a certain place*" and "*misinformation about result from mass community-wide COVID-19 testing in Wuhan*".

Access to and experience with COVID-19 tests

In the UK, few tweets (1.1%, 43/3856) shared their experience of taking a COVID-19 test, of which, 69.8% (30/43) reported being uncomfortable, 16.3% (7/43) reported being nervous, and 14.0% (6/43) reported no discomfort. Comparatively, there was a much wider report of the overall experience of taking a COVID-19 test from China (8.1%, 753/9299) and Beijing (7.0%, 221/3155). Furthermore, 9.2% (356/3856) of tweets in the UK concerned the logistical process of obtaining a test, including access to an appointment (7.0%, 25/356), wait time to take a test (including queues and heatstroke while waiting) (14.9%, 53/356), wait time for the test result (23.0%, 82/356), and others (59.3%, 211/356); a much wider reporting of the logistical process of obtaining a test were found from London (16.6%, 132/794). Similar discussions about the logistical process of obtaining a test were more common in China (21.6%, 2005/9299) and Beijing (23.9%, 754/3155), where wait time to take a test got major discussion (around 75%).

The price of COVID-19 testing was mentioned most commonly in London (6.7%, 53/794), and these discussions

included calling for free testing and dissatisfaction with high expenses in the UK, and expressed complaints about self-paying without reimbursement by medical insurance and satisfaction with free of charge testing in some places. For priority groups for COVID-19 testing, tweeters in the UK mentioned support for weekly testing for NHS staff and care staff and calling for priority testing to be extended. In China, posts concerning priority groups for COVID-19 testing referred to support for people with possible risk exposure, taxi drivers, and couriers to take priority testing.

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Table 1 Uptake experience of COVID-19 tests in the United Kingdom and China, N (%)

Themes	UK	London	China	Beijing
Total	**3856**	**794**	**9299**	**3155**
Uptake of COVID-19 tests				
1 Uptake of COVID-19 tests	178 (4.6)	39 (4.9)	3318 (35.7)	1462 (46.3)
1.1 Have a plan to take a test	35 (0.9)	7 (0.9)	811 (8.7)	417 (13.2)
1.2 Have taken a test	143 (3.7)	32 (4.0)	2507 (27.0)	1045 (33.1)
2 Reasons for taking a COVID-19 test	86 (2.2)	25 (3.2)	1600 (17.2)	824 (26.1)
2.1 Personal health needs	37 (1.0)	14 (1.8)	163 (1.8)	64 (2.0)
2.2 Mandatory testing policies for travel	30 (0.8)	9 (1.1)	659 (7.1)	223 (7.1)
2.3 Community-wide testing by governments	19 (0.5)	2 (0.3)	784 (8.4)	543 (17.2)
3 Other people taking COVID-19 tests	86 (2.2)	16 (2.0)	277 (3.0)	102 (3.2)
Sentiment towards COVID-19 tests				
4 Concern on effectiveness of COVID-19 tests	164 (4.3)	49 (6.2)	417 (4.5)	164 (5.2)
4.1 Trust tests to be effective	34 (0.9)	16 (2.0)	275 (3.0)	102 (3.2)
4.2 Doubt effectiveness of tests	130 (3.4)	33 (4.2)	142 (1.5)	62 (2.0)
5 Expiration date of COVID-19 tests	1 (0.03)	0 (0)	87 (0.9)	42 (1.3)
6 Incidental risks of COVID-19 tests	14 (0.4)	3 (0.4)	206 (2.2)	113 (3.6)
Perception of COVID-19 risk				
7 Perception of COVID-19 risk	152 (3.9)	45 (5.7)	813 (8.7)	299 (9.5)
7.1 High risk	139 (3.6)	44 (5.5)	789 (8.5)	286 (9.1)
7.2 Low risk	13 (0.3)	1 (0.1)	24 (0.3)	13 (0.4)
Inquiries and Rumours related to COVID-19 tests				
8 Inquiries about COVID-19 tests	75 (1.9)	21 (2.6)	517 (5.6)	137 (4.3)
9 Rumours related to COVID-19 tests	55 (1.4)	12 (1.5)	192 (2.1)	27 (0.9)
Access to and experience with COVID-19 tests				
10 Experience of taking a COVID-19 test	43 (1.1)	8 (1.0)	753 (8.1)	221 (7.0)
10.1 Feel uncomfortable	30 (0.8)	5 (0.6)	476 (5.1)	118 (3.7)
10.2 Feel nervous	7 (0.2)	2 (0.3)	234 (2.5)	70 (2.2)
10.3 Did not feel discomfort	6 (0.2)	1 (0.1)	111 (1.2)	43 (1.4)
11 Logistical process of obtaining a COVID-19 test	356 (9.2)	132 (16.6)	2005 (21.6)	754 (23.9)
11.1 Access to an appointment	25 (0.6)	8 (1.0)	93 (1.0)	52 (1.6)
11.2 Wait time to take a test	53 (1.4)	19 (2.4)	1579 (17.0)	557 (17.7)
11.3 Wait time for the test result	82 (2.1)	10 (1.3)	174 (1.9)	114 (3.6)
11.4 Others	211 (5.5)	99 (12.5)	186 (2.0)	48 (1.5)
12 Tribute to medical staff	69 (1.8)	15 (1.9)	1914 (20.6)	480 (15.2)
13 Price of COVID-19 testing	96 (2.5)	53 (6.7)	598 (6.4)	196 (6.2)
14 Priority groups for COVID-19 testing	665 (17.2)	9 (1.1)	271 (2.9)	101 (3.2)
Others	156 (4.0)	43 (5.4)	420 (4.5)	164 (5.2)

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Attitude and uptake of COVID-19 tests by characteristics of social media users

Table 2 shows the univariate analysis of attitude and uptake of COVID-19 tests by socio-demographic characteristics using Weibo data from China. Male Weibo users and those over 30-years-old were more likely to have a positive attitude towards individual COVID-19 testing ($p<0.001$), but male users were less likely to have a positive attitude towards mass community-wide COVID-19 testing led by the government ($p<0.001$). Additionally, females, users under 30 years old, and those with bachelor degree or above were more likely to take a COVID-19 test ($p<=0.001$), and there is no significant difference on reasons for taking COVID-19 test.

Table 2 Attitude and uptake of COVID-19 tests by characteristics for Chinese Weibo users

Variables	Attitude towards individual COVID-19 test, N (row%)				Attitude towards community-wide COVID-19 test, N (row %)			Uptake of COVID-19 test, N (row %)			Reasons for taking COVID-19 test, N (row %)			
	Total	**Positive**	**Neutral**	**Negative**	**Total**	**Positive**	**Negative**	**Total**	**Yes**	**No**	**Total**	**Personal health needs**	**Mandatory testing policies for travel**	**Community-wide tests**
Gender														
Male	2844	989 (34.8)	1587 (55.8)	268 (9.4)	660	614 (93.0)	46 (7.0)	2844	808 (28.4)	2036 (71.6)	413	37 (9.0)	167 (40.4)	209 (50.6)
Female	5940	1626 (27.4)	3814 (64.2)	500 (8.4)	910	882 (96.9)	28 (3.1)	5940	2470 (41.6)	3470 (58.4)	1170	126 (10.8)	487 (41.6)	557 (47.6)
Total	8784	2615 (29.8)	5401 (61.5)	768 (8.7)	1570	1496 (95.3)	74 (4.7)	8784	3278 (37.3)	5506 (62.7)	1583	163 (10.3)	654 (41.3)	766 (48.4)
p-value			**0.001**			**0.001**			**0.001**				0.440	
Age, years														
[10,30)	2705	680 (25.1)	1793(66.3)	232 (8.6)	356	344 (96.6)	12 (3.4)	2705	1187 (43.9)	1518 (56.1)	506	60 (11.9)	229 (45.3)	217 (42.9)
[30,90]	1125	420 (37.3)	602 (53.5)	103 (9.2)	269	251 (93.3)	18 (6.7)	1125	316 (28.1)	809 (71.9)	193	22 (11.4)	71 (36.8)	100 (51.8)
Total	3830	1100 (28.7)	2395 (62.5)	335 (8.7)	625	595 (95.2)	30 (4.8)	3830	1503 (39.2)	2327 (60.8)	699	82 (11.7)	300 (42.9)	317 (45.4)
p-value			**<0.001**			0.054			**0.001**				0.090	
Education														

Bachelor or above	2297	704 (30.6)	1387 (60.4)	206 (9.0)	472	450 (95.3)	22 (4.7)	2297	922 (40.1)	1375 (59.9)	444	45 (10.1)	166 (37.4)	233 (52.5)
High school or below	6582	1945 (29.6)	4067 (61.8)	570 (8.7)	1122	1068 (95.2)	54 (4.8)	6582	2395 (36.4)	4187 (63.6)	1162	118 (10.2)	493 (42.4)	551 (47.4)
Total	8879	2649 (29.8)	5454 (61.4)	776 (8.7)	1594	1518 (95.2)	76 (4.8)	8879	3317 (37.4)	5562 (62.6)	1606	163 (10.1)	659 (41.0)	784 (48.8)
p-value			0.491			0.897			**0.001**				0.159	

Boldface indicates statistical significance (p<0.05).

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Discussion

This infodemiology study assess social media posts in the UK (Twitter) and in China (Sina Weibo) to assess public attitudes and uptake experience around individual COVID-19 testing and government-led mass community-wide testing. Social media users in the UK demonstrated a higher percentage of support towards COVID-19 testing than those in China; support was higher across the UK and mainland China than it was in their respective capital cities, London and Beijing – from which most opposition originated. Comparatively, there were much wider reports of public uptake of COVID-19 tests in mainland China (35.7%) than the UK, and uncomfortable experiences and logistical barriers to testing were more expressed in China. In the UK, people particularly cared about priority groups of COVID-19 testing. In China, females, those under 30 years old, and those with a bachelor degree or above were more likely to have taken a COVID-19 test.

The analysis of social media data showed that, despite having a higher case count, a much smaller proportion of tweets in the UK (3.6%) expressed a high perceived risk from the COVID-19 pandemic, compared with that of mainland China (8.5%). The top reasons for taking COVID-19 tests were different in the UK and China: personal health needs due to possible risk exposure or symptoms were the leading reason in the UK, but community-wide testing or mandatory testing policies for travel were the leading reason in China. Furthermore, in both countries, the number of people who expressed neutral attitudes towards COVID-19 testing increased to become the majority over time. Scholars(19, 20) have argued that widespread dissemination of cheap and rapid tests might be as effective as a vaccine at interrupting coronavirus transmission by identifying and isolating people with the virus when they are most infectious. Our data showed that as the public became familiar with the concept of

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COVID-19 testing, they developed neutral attitudes towards testing as it became a normal part of life during the pandemic, which can lead to an increase in acceptance and uptake behaviour when it is needed. Policies and information campaigns that emphasise the importance of timely testing and provide a supporting environment for accessing a test are key to increasing uptake.

Accessibility, frequency, and sample-to-answer time were common topics of discussion on social media, and were identified as the most important factors for screening to be effective – even more of a priority than the analytical limits of detection (20). Also, inquiries and rumours related to COVID-19 testing pointed to a lack of frequent and correct information. These findings pointed to a need for effective emergency risk communication strategies during a public health crisis that are informed by evidence derived from ongoing social listening that not only meet public information needs but also debunk rumours and misinformation in real time.

Our findings showed that the local epidemic situation influenced public attitudes towards individual COVID-19 testing and mass community-wide testing, as well as highlighting the challenges for governments when seeking to balance epidemic control and livelihoods; it also emphasizes the importance of public communication about health risks and counter strategies. This study was performed when the UK was under its first nation-wide lockdown; at that time, China had lifted its nation-wide lockdown for more than two and a half months. tweets from London and the UK showed overwhelming support for both strategies, whereas more Weibo users expressed neutral or opposition attitudes toward mass community-wide testing in China. In the UK, the government has long been criticised for being under-prepared for the COVID-19 pandemic and therefore lacking testing capacity

for both the general public and frontline workers. Without reliable test results, there was very limited data available to develop and introduce an exit strategy for the general lockdown, as health experts had no evidence to inform their decisions. On the other hand, comprehensive testing requirements around domestic travel within China have been in place since March 2020, when the country resumed normal activities. Between 12 and 22 June 2020, the Beijing government led mass community-wide testing, completing 2.95 million tests in 10 days due to a small resurgence of cases in the community. As such, even though Chinese people generally reported a higher perceived risk toward COVID-19, some questioned the cost-effectiveness of implementing such mass testing when daily new cases in China fluctuated under 60. Moreover, we found few people discussed the duration (expiration) of test results and incidental risks associated with COVID-19 tests such as cross-infection and threat of asymptomatic infection from a crowd gathering. More risk communication efforts are needed to raise awareness.

There are geographic differences in public attitudes towards mass community-wide COVID-19 testing, with more people in London and Beijing and adjacent areas opposing testing. In China, strong evidence indicates gender differences in attitudes towards mass community-wide COVID-19 testing, with more female residents supporting government-led testing. The data indicates that the public is a diverse, heterogeneous set of people with varying experiences and needs and who react to messages differently based on their points of view. Tailoring engagement strategies to the target community will be critical in increasing acceptance towards COVID-19 testing and other containment measures.

There is an inherent bias shared among all social media studies, where users might present themselves differently online and/or represent a skewed-younger population(21). However, findings from this

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study demonstrated that monitoring social media data during an epidemic can support real-time assessment of public attitudes, updates, and experiences of key containment measures, inform communication and community engagement strategies, and assess the effectiveness of these strategies. To avoid the possible bias resulting from some Weibo posts being removed by the government, Weibo posts used for this study were downloaded daily for analysis before any interference.

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Declaration of interests

We declare no competing interests.

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